

Beryl Live Human OS — Realism, Consciousness & the JEV Integration

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1. One thesis, not two problems

Every failure this project has logged so far splits cleanly into two categories — **appearance** and **consciousness** — but they turned out to be the same failure wearing two faces.

On appearance: a CSS "life layer" scored 96/100 on the project's own Realness Index, and the director looked at it and said "*it is not alive — this is a blinking portrait.*" He was right. The layer was transform math on a flat photograph, and no amount of tuning gets transform math to close an eyelid.

On consciousness: the AWARENESS 3/4 pillar — situational self-knowledge, i.e. does she know what she actually is right now — scored **0**. The system was in stage L1. She said "*I'm at L0 right now.*" Not a rendering bug. A proprioception failure: her stated self-model disagreed with her actual telemetry, and nothing caught it.

The connecting thread, named plainly in the orchestration pivot: in both cases, the certification that was supposed to catch the failure didn't. The MOTION pillar read transform *values* instead of asking what was actually *painted*, and passed dead output twice — once while every depth layer animated perfectly *behind an opaque portrait*. A high score on the wrong metric is worse than no score at all, and that is exactly what happened on both appearance and consciousness: the evaluator was checking a proxy signal, not the ground truth it was supposed to stand in for.

That is the real problem this paper is about. Not "her face doesn't look real enough" and "her self-model is wrong" as two separate bugs to fix — but **the absence of an evaluator that can be trusted**, on either axis, running continuously enough and cheaply enough to actually catch the failure the moment it happens instead of a periodic certification script that can be fooled twice by the same kind of mistake. That absence is what JEV closes.

2. What JEV actually is

JEV is TypeSafe AI's first public release in a new model class they call **System One models** — named directly after Daniel Kahneman's *Thinking, Fast and Slow*. Existing LLMs, with chain-of-thought and sequential token generation, are System 2: slow, deliberate, expensive per call. A System One model is built for the other half of cognition — classify, route, score, extract, branch — and do it in one shot, not a conversation.

Architecturally, JEV is non-autoregressive: a parallel sampler produces the entire structured output in a single pass rather than generating text token by token. It does not write prose and then get parsed — it returns a typed decision plus a calibrated confidence score directly, constrained to a schema you define. Because the output space is constrained by construction, TypeSafe reports a **0% structured-output error rate** and **0% tool-call error rate** — not "rarely fails validation," but structurally cannot emit an invalid value.

Training uses a method TypeSafe calls Reinforcement Learning for Calibrated Decisions (RLCD), which optimizes for *epistemically honest* probabilities rather than human preference: higher stated confidence is trained to actually mean higher accuracy. This matters more than it sounds — it's the difference between a model that sounds sure and a model whose sureness is calibrated to be trustworthy.

The numbers, reported honestly, cost included:

	JEV	Comparable LLM judge
Latency	70–500ms (avg ~0.4s)	10–38s
Cost	\$0.042 / million input tokens, output free	e.g. \$2.00/M for GPT-5.6 Terra
Structured output error rate	0% (by construction)	nonzero
Judge-score variance (LangSmith agent-eval test)	—	Jev's variance measured 92–913× <i>lower</i> than GPT-5.6 Luna/Terra and Claude Sonnet 4.6 — i.e. far more repeatable
Cost on a real 5-example eval set (LangSmith)	\$0.34 total	\$28.17 (Claude Sonnet 4.6)
Accuracy, TypeSafe's own 4-workflow benchmark	67.8%	GPT-5.6 Terra: 67.9% — essentially tied

The honest tradeoff, stated plainly: JEV is not more accurate than a frontier LLM judge. On TypeSafe's own benchmark it's a statistical tie with GPT-5.6 Terra. What it trades peak accuracy for is roughly 193× the speed and 445× the cost efficiency, with far higher repeatability. That is precisely the profile of a **guardrail**, not a **judge of last resort** — something built to run on every single call, not something built to be right about hard cases. TypeSafe's own framing agrees: System One models are aimed at decisions *inside* software — verification, routing, scoring — not at chat, and not at replacing deep reasoning.

Status, as of today: early access, behind a waitlist, as of mid-September 2026. Reached via POST `https://api.typesafe.ai/v1/systemone`, model route `jev-latest`, official Python and JS SDKs. The architecture is unpublished and weights are not released — this is a proprietary API dependency, not an open-source component. That distinction matters for section 7.

3. Why this is the missing piece, not just a nice-to-have

The existing architecture already has a two-speed design that nobody had a model for. L0 (idle presence) has a hard budget: **control plane answers in under 1 second, always**. L1 and L2 are allowed to be slower — A2F, Omniverse cinematic takeover, deliberative escalation. The original plan for the orchestration layer was Microsoft Agent Framework's

Magentic pattern: an LLM-based supervisor manager deciding, on every turn, who acts next and why. That is a System 2 decision-maker, and it was being asked to also make System 1-shaped decisions — is this frame valid, is this node lying about its state — at LLM latency and LLM cost, inside a budget that does not have room for either.

JEV fits the L0 budget the Magentic supervisor structurally cannot. At 70–500ms and a fraction of a cent per call, a JEV classifier can run on every frame or every utterance, not a sampled certification pass. This is the direct fix for two named, specific failures:

- **The MOTION pillar, fixed at the root cause.** The certification failed twice because it read transform *values* as a proxy for what was painted. A JEV classifier wired to the actual rendered frame — not the CSS transform driving it — with a schema like `{is_motion_visible: bool, confidence: float, occluding_layer: string | null}` cannot be fooled by a value that looks right on paper, because it never sees the value — it sees the pixels, every frame, for a third of a cent per thousand frames.
- **AWARENESS 3/4 (proprioception), fixed structurally instead of caught after the fact.** The failure was a mismatch between her stated stage ("I'm at L0") and her actual telemetry (L1). A JEV call with schema `{stated_stage: string, telemetry_stage: string, consistent: bool, confidence: float}`, run on every utterance against `/v1/nodes/{id}` output, catches this the instant it happens — not on the next scheduled certification run.

The two-speed mapping is precise, not a rebrand of existing plans:

Layer	Budget	What runs there	Why
L0 — always-on verification	<1s, every frame/utterance	JEV — typed, calibrated, cheap enough to never sample	Matches System 1: fast, constrained, high-volume
L1/L2 — diagnosis and dialogue policy	Seconds, on escalation	Magentic / Handoff / Group Chat (unchanged from the pivot doc)	Matches System 2: deliberative, needs real reasoning, run rarely enough that its cost is fine

JEV doesn't replace the orchestration plan from the pivot doc — it fills the layer underneath it that the plan never had an answer for: the always-on, structurally honest verifier that decides *when* something is wrong cheaply enough to check constantly, so

the Group Chat diagnosis quorum only has to convene when there's a real problem to reason about, not to catch things a script should have caught.

4. Endless possibilities — each one a mechanism, not a slogan

"Endless possibilities" is the brief, but every item below is grounded in a specific schema, a specific node, or a specific number from section 2 — not aspiration.

Live certification instead of periodic certification. `scripts/ecc5-certify.mjs` currently runs on demand and produces a snapshot — which is exactly how a placeholder got scored 96/100 and how MOTION passed dead output twice. At \$0.042/M input tokens and sub-500ms latency, running the same class of check on every frame or turn instead of on a sampled run stops being a cost question. The ECC-5 certificate becomes a live number instead of a report card.

A cross-check on SoulX-Duplug's listening state. Duplug already classifies audio into idle/nonidle/speak/blank in real time — that's the raw signal. A JEV call with schema `{duplug_state: string, audio_energy_pattern: string, agrees: bool, confidence: float}` gives that raw classification a second, independently-trained, calibrated opinion — cheap enough to run on every turn, which turns "conversational awareness" from a single classifier's output into a verified one.

An always-on fault-injection auditor. The pivot doc's standing rule is that fault injection must hit the real path — shifting audio against Pipe 3's text-derived visemes should produce a *null* result, not a fake desync. A JEV classifier watching for exactly that null-result signature turns "we tested this once and wrote it down" into a check that runs every time the pipeline is touched, catching the day someone accidentally rewires Pipe 3 to derive visemes from audio instead of text.

Typed routing inside the Handoff graph. LangChain's own production pattern — a "Jev Classifier" making routing and tool-call decisions inside an agent's control loop — maps directly onto the Handoff pattern's node-to-node edges (ASR → Agent → TTS → A2F). Not every handoff needs an LLM to decide where a signal goes next; many are classification problems JEV is built for, freeing the Magentic supervisor to spend its (expensive, slow) reasoning only on handoffs that are actually ambiguous.

A liveness guardrail on the render pipeline itself. Whichever render stack ships — FlashHead, LeapTalk, or the pipeline still being chosen — a JEV classifier can watch its output stream directly: `{frame_has_motion: bool, occlusion_detected: bool,`

`blink_in_range: bool}`, run continuously, as the thing that would have caught the opaque-portrait failure in real time rather than in a certification run written after the fact.

Evaluation as a shipped product feature, not a research cost. TypeSafe's own LangSmith comparison ran five test examples for \$0.34 against \$28.17 for an LLM judge — a $\sim 83\times$ gap on a toy set that only widens at product scale. That is the difference between "we ran a certification before the demo" and "every user's session is verified live," and it's a genuine product capability, not just cheaper QA.

5. What makes this ours

The prior survey of the field — LiveAvatar, Hallo-Live, SoulX-FlashHead, LeapTalk, Avatar-Forever, and everything else checked against source — published generation-quality metrics: FPS, FID, sync-C, latency to first frame. **Not one of them publishes a runtime self-verification layer.** They report how good the model was in evaluation; none of them ship a mechanism for verifying, live, in production, on a specific user's session, that what's being rendered right now is actually what it claims to be.

That gap is exactly where Beryl Live Human OS can differentiate, and it isn't a marketing claim — it's the literal shape of the two failures already logged. Nobody else in the surveyed field has had a certification score a placeholder 96/100 in public and had to explain why, because nobody else in the surveyed field runs a certification against their own output at all. Owning that — a photoreal, license-clean, cost-bounded render stack *plus* a live, calibrated, structurally-honest verification layer wired into every frame — is a combination nothing in the survey has, not because it's exotic, but because nobody building the render side has needed an answer to "how do you know she's actually alive right now" the way this project was forced to by its own two public failures.

The brand case, stated plainly: everyone else is shipping a face. Beryl Live Human OS would be the one shipping a face **with a receipt** — a verifiable, per-session, cheap-enough-to-run-always answer to the exact question the director asked in August: *is this actually alive, or does it just look alive on a chart.*

6. On "consciousness" — precisely, not loosely

This paper uses "consciousness" the way the project's own certification does: as a name for **situational self-knowledge** — does the system's stated model of itself match what's actually true of it — not as a claim that anything here is sentient or has subjective experience. AWARENESS 3/4 failing because the system said "I'm at L0" while telemetry

showed L1 is a **proprioception** failure, in the same sense a person misjudging where their own arm is has a proprioception failure. It is a real, measurable, fixable property. It is not a claim about inner experience, and nothing in this paper should be read as making one.

What makes JEV unusually well-suited to this specific, narrow, honest definition is its training objective: RLCD optimizes for *epistemically honest* probabilities — a model whose stated confidence is trained to actually track its accuracy. That is close kin to what proprioception verification needs: not a model that sounds certain, but a model whose certainty means something. Using JEV to check "does her self-report match her telemetry" is not a claim that JEV understands consciousness — it's a calibrated classifier checking a well-specified, falsifiable, boring fact: do these two numbers agree. That boring fact is exactly what was missing when AWARENESS 3/4 scored 0, and closing that gap is the actual, buildable claim here — appearance of coherent self-knowledge, verified continuously, not consciousness itself.

7. Honest caveats

- **The API key could not be verified in this environment.** This session's environment variables were checked directly and contain nothing matching TypeSafe or JEV. A deeper filesystem search for credential files was blocked by this session's own security guardrail (a "credential exploration" classifier) and was not worked around — that block is a legitimate safety boundary, not a bug to route past. If the key was exported on your laptop rather than this cloud session, or in a shell profile this container doesn't source, it will not be visible here. This should be confirmed directly against `api.typesafe.ai` before any integration work assumes it's live.
- **JEV is proprietary, not open source.** Architecture unpublished, weights not released, API-only, currently in early access behind a waitlist. This sits outside the license-integrity framework established in the earlier report (the Apache-2.0, dependency-audited FlashHead/LeapTalk/Duplug stack) in a specific, tractable way: called as an HTTP API rather than vendored as code, it doesn't attach license obligations to the codebase the way the Gemma or Stable Audio dependencies would have — but it is a new vendor dependency, with its own availability and pricing risk, and belongs in the same dependency inventory as everything else, tracked as a service dependency rather than a code dependency.
- **JEV is not more accurate than a strong LLM judge** — 67.8% versus GPT-5.6 Terra's 67.9% on TypeSafe's own benchmark, a statistical tie. Its case rests entirely on being ~193× faster, ~445× cheaper, and far more repeatable (92–913× lower variance), which is the right tradeoff for a guardrail that must run on every call, and the wrong tool for a

hard diagnostic judgment call that runs rarely. Every use case in section 4 was chosen because it's a narrow, well-specified, high-frequency check — not because JEV should replace the Magentic/Group-Chat diagnosis layer, which still needs real deliberative reasoning.

- **Early access means availability risk.** A waitlisted, mid-September-2026 product is not yet a dependency to build a critical path on without a fallback — the L0 verification layer described in section 3 should degrade gracefully to the existing certification script if the API is unavailable, not hard-fail the control plane.
- **This paper is a design direction, not a shipped integration.** Nothing here has been tested end-to-end against the actual `ace-controller` telemetry or a real render pipeline. The next concrete step is a small proof of concept: one JEV call, one schema, wired to one real signal — most usefully the AWARENESS 3/4 proprioception check, since it's the one pillar with a documented, reproducible failure to test against.

8. Plan A, locked 2026-09-21

This is now Plan A, authoritative until a specific, evaluated alternative demonstrably beats it — not superseded by a new model being mentioned, only by one that's been checked against source the way everything here was.

- **Render:** LeapTalk (Apache-2.0), forked on SoulX-FlashHead-1.3B, on 1× RTX 4090.
- **Turn-taking:** SoulX-Duplug 0.6B (Apache-2.0), listening state feeding the brain.
- **Foundation:** Nemotron Agent (existing) + a new Advanced Memory layer, cross-session episodic/vector recall.
- **Verification layer:** as of today, **self-hosted Kev is primary**, not JEV — the reasoning is in section 9. JEV remains an available hosted upgrade path for the specific cases where its higher calibration ceiling is worth the API cost and the proprietary dependency.

Everything else from the prior sections — the license audit, the NOTICE-file practice, the CI dependency scan, the fork provenance table — still applies unchanged. Locking Plan A doesn't mean stopping the search; it means the bar for replacing any piece of it is now "checked against source and demonstrably better," not "looked promising in a search result."

9. Kev — the open-source verification layer

Correction on provenance, stated plainly because it matters for section 10: `tyronne-os/kev-beryl-brain` is not original work yet. Checked directly — every commit in the visible history is authored by Jared Palmer (`jp@ cognition.ai` , the Formik/TSDX/Razzle author, now at Cognition), the last one dated today. This is his `jaredpalmer/kev` project, copied into this account. That's a completely normal starting point for a fork — the same practice this whole research thread has been built around — but it needs to be named correctly, not presented as Beryl's own research, or the "owning the science" brand claim in section 10 collapses the first time someone checks `git log` .

What Kev actually is: an open reproduction of Jev's architecture (credited explicitly to a third-party write-up, "*Jev's Architecture Unmasked*"), built on Qwen3.5 bases (0.8B / 4B / 9B, plus a previous Qwen3 generation kept for Apple Silicon latency), trained with LoRA + a pointer head that scores option tokens rather than generating text — the same non-autoregressive, typed-decision shape as Jev. **It speaks JEV's own API:** `POST /v1/systemone` , and TypeSafe's official Python SDK works against it unmodified by pointing `base_url` at your own server. The three schemas designed for the pipeline prototype — proprioception, motion, listening cross-check — run against Kev with no changes.

Fully open, license-clean, matching the standard already set: Apache-2.0 for the code and for the Qwen3.5 base models. Training datasets carry their own licenses, disclosed per-checkpoint in the model cards rather than buried — the same transparency this project's own NOTICE-file practice was built to enforce, which is a good sign about the project's own discipline.

Self-hosted, so the API cost objection is not just reduced, it's zero. `uv run python -m kev.serve --run jaredpalmer/kev-4b --port 8009` runs on CUDA, ROCm, or Apple Silicon. On CUDA with `flash-linear-attention` installed, a five-question request takes **tens of milliseconds on an H100** — comparable to or faster than JEV's own 70–500ms, and the marginal cost per call is your own GPU-second, not \$0.042/M tokens paid to a vendor.

The honest gap, reported by the project itself, not hidden:

	Kev-9B	Jev
Accuracy, new sources (dev/test)	0.822 / 0.852	0.857 / —
Brier score, new sources (lower is better)	0.286 / 0.237	0.211 / —

	Kev-9B	Jev
Share of decisions automatable at a 5% error budget	0.45-0.57	0.70
Confident wrong answers (≥ 0.9 probability, new sources)	4.0%	3.7%

Kev-9B trails Jev by roughly 3.5 points on out-of-distribution accuracy, and its calibration ceiling — the share of decisions you can safely automate without a human check — is meaningfully behind. The project's own README says it plainly: *this isn't a controlled comparison, since nobody knows what Jev was trained on*. That honesty is itself evidence Kev is a credible thing to build on — a repo that oversold its own numbers would be a worse foundation regardless of the numbers themselves.

What this does and does not unlock — scoping the premise in the question. Kev replaces JEV specifically in the verification mesh: the proprioception check, the motion check, the listening cross-check. It does **not** touch, and does not give "open-source freedom to improve," the render nodes — LeapTalk, FlashHead, and Duplug remain separate Apache-2.0 projects with their own separate codebases and their own separate fine-tuning paths. One repo does not unlock all major nodes; it unlocks the one node this whole research thread identified as the actual gap.

The real unlock, and it's a real one: `kev.train --init_from jaredpalmer/kev-4b` warm-starts a fine-tune from the released checkpoint on your own labeled examples — a few hundred are enough, per the project's own documented case study (a from-scratch fine-tune scored 0.33 on a downstream eval; the same data starting `--init_from` the released checkpoint scored 0.88). Fine-tuned on Beryl's own MOTION-pillar dead-output cases and AWARENESS 3/4 proprioception mismatches — the two failures that started this entire research arc — that becomes a decision model nobody else has, because nobody else has that training data. That, not the base fork, is the actual path to owning something here.

One operational flag, unrelated to licensing: the server binds `127.0.0.1` with **no authentication by default**. Fine for local development, not fine to expose without adding auth first.

Verdict: run Kev-9B or Kev-4B self-hosted as the primary verification model — free, fast on CUDA, license-clean, and fine-tunable on proprietary data. Keep JEV available as a fallback for anything where the ~3.5-point calibration gap genuinely matters more than the cost and dependency, once its API key is actually confirmed live.

10. Owning the science — a brand identity direction

Every brand claim below is load-bearing on something already built or already found true in this research — not aspiration dressed as strategy. And the Kev provenance finding above sets the one hard rule for all of it: **the brand can only claim what's actually been done**. Forking is normal; presenting a fork as original research is the one move that would break this identity the first time anyone checks.

The throughline, and it's real: this project has now had two public failures — a certification that scored a CSS placeholder 96/100, and a proprioception pillar that scored 0 because the system's self-report disagreed with its own telemetry — and both times, the response was to name the failure precisely and build a structural fix, not a better demo. Nobody else in the entire field surveyed across this research publishes anything like that. They publish FID scores and FPS numbers. Not one of them publishes a runtime self-verification layer, and not one of them has had to explain, in public, why their own evaluation gave a wrong answer twice.

Brand pillar 1 — the receipt, not the reel. Every claim about realism ships with the live certificate that proves it, the same way ECC-5 already works: a number, a method to reproduce it, and an honest gate ("back to the lab" is a real, sayable outcome, not something a brand hides). Competitors show a polished clip. This shows the clip *and* the proof.

Brand pillar 2 — calibrated, not confident. This is not a slogan invented for marketing — it's literally what JEV and Kev are trained to do: RLCD optimizes for probabilities that mean what they say, not for sounding sure. A brand voice that states calibrated uncertainty ("87% confident, here's why") instead of unqualified certainty is a genuinely rare register in an industry that oversells by default, and it costs nothing extra to say, because the system underneath is already built to produce exactly that number.

Brand pillar 3 — fork in the open, own the delta. The NOTICE-file practice, the CI license scan, and now the `--init_from` fine-tuning path aren't just legal hygiene — made visible, they're the R&D culture: publish what was forked and from whom, publish what's proprietary and why, never blur the two. A public `NOTICE.md` crediting LeapTalk, FlashHead, Duplug, and Kev by name — the way this section credits Jared Palmer by name — is a trust signal competitors building on the same open-source substrate mostly don't bother making visible.

Visual system — promote what's already operational, don't invent new marketing colors. The product's own engineering palette already encodes the split this brand is about: electric blue for the living system (presence, motion, conversation — the

render/brain/memory path), gold for the proof layer (research badges, the JEV/Kev verification annotations, already used exactly this way in both the pivot doc's node graph and the pipeline prototype). Turning that internal convention into the public brand system is unusually credible precisely because it wasn't designed as marketing — it's what the engineers already reach for when they need to mark "this is verified."

Naming, one confirmed and one aspirational — kept separate on purpose. "Beryl Live Human OS" as the master brand is already in use in this research and fits: it names the platform, not a single avatar. "ECC-5" as a public-facing trust mark — something other builders in this space could eventually cite the way video work cites a codec or a color standard — is a real possibility given the certificate already exists and already fails things honestly, but it's aspirational: it requires the certification to survive contact with outside scrutiny first, which hasn't happened yet. State it as a direction being worked toward, not a claim already earned.

11. The Clay — Datasets, Benchmarks, and the Path to FaceTime Realism

The FaceTime-friend standard separates into five measurable dimensions — emotion, micro-expression, gaze, hair physics, and body micro-movement. Each has publicly proven datasets, benchmarks, and training milestones already accumulated by academic labs. None of this requires Kev to become a new model class; it requires Kev's `--init_from` fine-tuning pipeline trained on the right signal for each dimension, then wired to the right driving input in the render layer.

Dimension 1 — Emotion & Expression. Three datasets cover this cleanly: **AffectNet** (450K+ images, 8 basic + 11 compound categories, continuous Valence/Arousal labels, MIT license — the most-cited benchmark in 2025–2026 expression papers); **BP4D+** (spontaneous 3D facial expressions, AU intensity-labeled, physiological signals co-registered — gold standard for fine-grained work); **MAFW** (10,045 video clips, multi-label compound emotion in live conversation, Apache-compatible). Fine-tune Kev-4B with `--init_from` on AffectNet + MAFW with schema `{emotion_label: string, valence: float, arousal: float, intensity: float, confidence: float}`. Every turn produces a typed emotion prediction. FlashHead and LeapTalk both support conditional expression driving — the predicted emotion feeds as a conditioning signal replacing the neutral default. The render stack doesn't change; the input it receives does.

Dimension 2 — Micro-Expressions. The datasets: **CASME II** (247 sequences of spontaneous micro-expressions, AU-coded, onset/offset labeled); **SAMM** (159 sequences, 26 subjects, FACS-coded, best for temporal precision); **MMEW** (300 clips, partial-face recognition). Micro-expressions are the hardest realism dimension because they're temporal — 1/25 to 1/5 of a second, barely perceptible consciously but registered as “alive” at a gut level. Kev's schema is a trigger: `{trigger_microexpression: bool, au_code: int, onset_delay_ms: int, duration_ms: int, confidence: float}`. The model predicts from dialogue context whether a micro-expression should appear and which action unit drives it. The render layer fires a timed AU injection. No micro-expression at all is what makes most current avatars read as hollow — not wrong, but absent.

Dimension 3 — Gaze & Blink. ETH-XGaze (110,856 images, 80 subjects, high-resolution gaze ground truth) and MPIIFaceGaze (in-the-wild, 45K images) cover estimation. Audio2Face-2D already exposes blink and gaze natively — this dimension is partially shipped. Kev adds semantic gaze: `{gaze_target: string, should_break_gaze: bool, blink_trigger: bool, confidence: float}` — the difference between eyes that move procedurally and eyes that break contact when a topic gets uncomfortable, or hold steady when making a point.

Dimension 4 — Hair & Soft Tissue. This is a render-layer problem, not a Kev problem. Kev's role is to predict head-movement magnitude that triggers physics responses downstream. The relevant work: NeuralHDHair (2022, neural strand-level hair, MIT license); HairStep (2023, text-driven strand control); PhysX strand solver already embedded in Omniverse. Kev schema: `{head_angular_velocity: float, direction: string, trigger_hair_simulation: bool, intensity: float}`. Hair that follows head turns with ≤ 2 frame lag is one of the most immediately perceptible realism signals current portrait avatars universally lack.

Dimension 5 — Body Micro-Movements. The datasets: MAAD (Micro-Action Dataset, 2024, 7,392 video clips, 77 micro-action classes including shoulder adjustments, neck tensions, breathing rhythm, finger movements); BEAT (Body-Expression-Audio-Text, 2023, 60 speakers $\times$ 34 emotions, synchronized body language + speech, Apache-2.0). The FaceTime-level realism test is not gross motion — it is listener behavior: the slight forward lean when something gets interesting, the micro-nod while the other person is speaking, the shoulder relaxation signaling comfort. Kev schema: `{listener_state: string, nod_trigger: bool, lean_direction: string, shoulder_tension: float, breathing_visible: bool, confidence: float}`. Fine-tuned on BEAT with `--init_from kev-4b`, this becomes an always-on body-language predictor at zero marginal cost and sub-50ms latency.

Training Milestones — a proposed sequence.

M1 — Emotion Baseline. Fine-tune Kev-4B on AffectNet + MAFW. Target: Valence/Arousal prediction accuracy ≥ 0.85 on held-out conversation clips. Gate: emotion-driven FlashHead output passes a human A/B test against neutral-expression baseline.

M2 — Micro-Expression Trigger. Fine-tune M1 checkpoint on CASME II + SAMM. Gate: spontaneous micro-expressions appear on ≥ 1 in 8 turns without being identified as artificial in a blind viewer study.

M3 — Hair Physics Coupling. Wire Kev head-velocity schema to Omniverse strand solver. Gate: hair follows head turns with ≤ 2 frame lag in Kev-triggered L1/L2 mode.

M4 — Body Language. Fine-tune M2 checkpoint on BEAT dataset. Gate: avatar produces listener nods and shoulder behavior that a blind rater scores $\geq 4/5$ on a “feels like a real person listening” scale.

M5 — Emotional Arc Continuity. Fine-tune M4 on IEMOCAP multi-turn sequences. Gate: Valence/Arousal trajectory stays contextually consistent across ≥ 5 consecutive turns without resetting to neutral.

What `--init_from` actually makes possible. The documented case study is stark: from-scratch fine-tune scored 0.33 accuracy; same data starting `--init_from` the released 4B checkpoint scored 0.88. That gap is what makes this milestone sequence tractable — each stage adds a few hundred labeled examples to a checkpoint that already understands calibrated decision-making. M2 builds on M1, M4 builds on M2. Each milestone’s training data is the proprietary moat: CASME II + SAMM + BEAT + the pipeline’s own proprioception and motion failure cases, assembled nowhere else. The base model exists, the fine-tuning infrastructure is documented and tested, the datasets are real and open. The milestones above are the path from a talking portrait to something that reads as a friend on a call.